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Abstract—

I. INTRODUCTION

the detector would still output prediction, but it is not safe in those condition we didnt train for. so it is also important to know when should we stop, apart from having the high classification and localization accuracy from the detector.

we found that, The earlier stage is more sensitive to the corruption in the image than in the later stage, which is more tuned for the detector output.

different corruption have different effect on the channel reaction. For example,... (example a bit why, some are increasing the average reaction, some are making the variance bigger)

At last, we were using the first stages to calcualte the channel behavior, compare them with the closest images in the cleaning image bank (selected with the later stage, which is more about he scenario setting of the image, because it doesnt seem reasonable to compare a beach image with some busy road image for example, they definitely have very different channel reaction, and that's why we choose to use closer images.)

We validate the result for different detectors... (some numbers and trend)

II. RELATED WORK

A detector that is accurate on its test set can still fail on the road, where it meets objects and conditions it was not trained for, and it keeps producing detections all the same. Such failures, which the safety of the intended functionality (SOTIF) addresses [1], call for a runtime monitor that tells the vehicle when to stop trusting its detector [2], [3]. Monitors differ in the question they answer. Out-of-distribution (OOD) detection asks whether the input contains an object of a class the detector was never trained on [4], [5]. Uncertainty estimation and failure prediction ask how far a given prediction can be trusted [6]–[9]. Covariate-shift detection [10] asks whether the input itself has changed while its content stays familiar, as when the image is foggy, blurred, noisy, or compressed [11], [12]. Our work belongs to this third kind. A monitor of this kind can take its signal from one of three places: the detector's predictions (Sec. II-A), the image (Sec. II-B), or the detector's features (Sec. II-C).

A. Monitoring the Prediction

Ideally, a model should be less confident on inputs it is not familiar with [13]. The detector's outputs are therefore a natural place to start. SAOD [14] averages the uncertainty of the three most confident detections to decide whether to accept an image. ContrastiveConf [15] computes the margin between the confidence of the queries a DETR keeps and

that of the queries it discards. Apart from the raw confidence, scores computed from the logits, such as the energy [16] and the maximum logit [17] are also used.

B. Monitoring the Image

The image itself can be judged by a model beside the detector. Image-quality assessment (IQA) models predict how degraded an image looks. They differ in what they learn from: clean images, degraded images, or neither. Models that learn from clean images flag departures from them. NIQE [18] measures how far an image's local contrast statistics depart from those of clean photographs. DisCoPatch [19] trains a discriminator to reject the imperfect patches of its own variational autoencoder, and flags an image whose patches the discriminator rejects. Models that learn from degraded images either recognize particular conditions from labeled examples, such as soiling [20], fog [21], or rain and snow [22], or learn one quality score from synthetically distorted images, as BRISQUE [23] and ARNIQA [24] do. CLIP-IQA [25] learns from neither: it rates an image zero-shot, by comparing it with the prompts "Good photo" and "Bad photo.". All of these run a second model beside the detector and leave unused the features the detector has already computed.

C. Monitoring the Detector's Features

Finally, a monitor can take its signal from the detector's internal features. Hashemi et al. [26] fit a mean and spread to each neuron on clean images, and flag an image when too many neurons leave mean $\pm$ two standard deviations. Becker et al. [27] compare each channel's activation distribution in a frozen detector with the channel's distribution over the training set. A learned alternative trains a small network on the detector's backbone features to embed how an image is degraded, using synthetically degraded training images, and flags an image whose embedding lies far from that of clean images [28].

Like Hashemi et al. and Becker et al., our monitor needs only the frozen detector and clean images. Instead of one global clean reference, however, it judges an image's early channels against clean scenes of the same content, retrieved by the detector's deepest stage, the one least affected by corruption. It also separates two effects of corruption on these channels: a flattening, where their strongest responses fall, and a re-weighting, where their levels move up and down.

III. METHOD

IV. EXPERIMENT

V. CONCLUSION AND LIMITATION

In summary, we developed a monitor for answering the question "when is the detector not reliable anymore?".

It is usable for almost all the detector, not constraint within one type of detectors.

It has better performance (some numbers)

However, as we already discussed in sec xx, the model are trained to produce similar activation under xx from the augmentation. So even though their performance might drop, the activation cannot give us much cues, and the monitor cannot work with those yet.

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